

Níckolas de Aguiar Alves (he/him)

 PhD candidate in physics

 Federal University of ABC

 Santo André, São Paulo, Brazil

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Education

2023–
Sep.

Doctor in Physics

Federal University of ABC (UFABC), Santo André, São Paulo, Brazil

Advisor: André G. S. Landulfo 

2021–2023
Feb. Apr.

Master in Physics

Federal University of ABC (UFABC), Santo André, São Paulo, Brazil

Advisor: André G. S. Landulfo 

Thesis: *Nonperturbative Aspects of Quantum Field Theory in Curved Spacetime*

2017–2020
Feb. Dec.

Bachelor in Physics

Institute of Physics, University of São Paulo (IFUSP), São Paulo, São Paulo, Brazil

Advisor: João C. A. Barata 

Appointments

2026–2027
Aug. Aug.

Visiting PhD Candidate

Scuola Normale Superiore di Pisa (SNS), Pisa, Toscana, Italy

Supervisor: Guilherme L. Pimentel 

Funded by the São Paulo Research Foundation (FAPESP) Grant No. 2026/02509-8

Publications

Research Articles

- 1 N. Aguiar Alves and A. G. S. Landulfo. *Sound as a gauge theory and its infrared triangle*. 2026. arXiv: 2512.15796 [hep-th]. Submitted.
- 2 N. Aguiar Alves and A. G. S. Landulfo. “Null infinity as a Killing horizon”. *Phys. Rev. D* **112**.6, 065017 (2025). arXiv: 2504.12514 [gr-qc].
- 3 N. Aguiar Alves, A. G. S. Landulfo, and B. A. Costa. “Positive mass in general relativity without energy conditions”. *Phys. Rev. D* **111**.4, 044027 (2025). arXiv: 2408.00154 [gr-qc].

Review Articles

- 1 N. Aguiar Alves. “Lectures on the Bondi–Metzner–Sachs group and related topics in infrared physics”. *Eur. Phys. J. C* **86**.5, 507 (2026). arXiv: 2504.12521 [gr-qc].

Pedagogical Articles

- 1 C. C. Rodrigues Evangelista and N. Aguiar Alves. “Using thermodynamics to learn gravitational wave physics”. *Eur. J. Phys.* **47**.2, 025604 (2026). arXiv: 2602.21261 [gr-qc].

Essays and Short Contributions

- 1 N. Aguiar Alves. “Physics as a branch of poetry”. *Am. J. Phys.* **94**.7 (2026), p. 507.
- 2 N. Aguiar Alves and B. A. Costa. *The Measure of a Mass*. 2025. arXiv: 2503.18963 [gr-qc]. Pre-published.

Theses

- 1 **N. Aguiar Alves.** “[Nonperturbative Aspects of Quantum Field Theory in Curved Spacetime](#)”. MSc thesis. Santo André, Brazil: Federal University of ABC, 2023. xxiv, 152 pp. arXiv: [2305.17453 \[gr-qc\]](#).

Other Works

- 1 **N. Aguiar Alves.** “[Quantum Field Theory in Curved Spacetime. An Introduction](#)”. Unpublished lecture notes. 2023.

In Preparation

- 1 **N. Aguiar Alves**, A. G. S. Landulfo, and A. D. Pereira. *Nonperturbative renormalization group flow for a particle detector*. In preparation.
- 2 **N. Aguiar Alves**, C. C. Rodrigues Evangelista, G. J. Olmo, and D. Rubiera-Garcia. *Platypus stars: Exotic compact objects supported by vacuum pressure*. In preparation.

Funding

2026–	📖	PhD Scholarship – Research Internship Abroad Explorations with Twistors and Cosmological Correlators São Paulo Research Foundation (FAPESP) Grant No. 2026/02509-8 Advisor in Brazil: André G. S. Landulfo  Supervisor abroad: Guilherme L. Pimentel 
2025–	📖	PhD Scholarship The Sky as a Killing Horizon and Other Topics on the Infrared Structure of Gravity São Paulo Research Foundation (FAPESP) Grant No. 2025/05161-0 Advisor: André G. S. Landulfo 
2023–2025	📖	PhD Scholarship Coordenação de Aperfeiçoamento do Pessoal de Ensino Superior (CAPES) Granted to pursue my PhD after placing first in UFABC's admission ranking
2021–2023	📖	MSc Scholarship The Functional Renormalization Group in Quantum Field Theory in Curved Spacetimes São Paulo Research Foundation (FAPESP) Grant No. 2021/07372-7 Advisor: André G. S. Landulfo 
2021	📖	MSc Scholarship Coordenação de Aperfeiçoamento do Pessoal de Ensino Superior (CAPES) Granted to pursue my MSc after placing first in UFABC's admission ranking
2019–2020	📖	BSc Scholarship Hyperbolic Equations São Paulo Research Foundation (FAPESP) Grant No. 2019/12158-4 Advisor: João C. A. Barata 

Academic Service

Conferences Organized

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|------|---|---|
| 2026 | 📖 | Thermality in Quantum Field Theory in Curved Spacetimes
Main organizer
Promoted by the ICTP South American Institute for Fundamental Research |
| 2023 | 📖 | Golden Wedding of Black Holes and Thermodynamics
Main organizer |

Academic Service (continued)

Schools Organized

- 2024–  [São Paulo School in Gravitational Physics](#)
Main organizer
Part of [ICTP Physics Without Frontiers](#) (second edition onward)
- 2018–2021  [Jayme Tiomno School in Theoretical Physics](#)
Founder and organizer of the first three editions

Peer Review

- 2026–  Reviewer for the [American Journal of Physics](#)

Scientific Societies

- 2025–  [Brazilian Physical Society](#)
- 2024–  [International Society for Quantum Gravity](#)

Teaching Experience

Minicourses and Workshops

- 2025  Introduction to Black Hole Physics
[II São Paulo School in Gravitational Physics](#)
Minicourse for early undergraduate students
- 2024  Infrared Symmetries of General Relativity
[I São Paulo School in Gravitational Physics](#)
Minicourse for advanced undergraduate and early graduate students
- 2023  Quantum Field Theory in Curved Spacetime
[Golden Wedding of Black Holes and Thermodynamics](#)
Minicourse for early graduate students
- 2021  Algebraic Methods of Theoretical Physics
[III Jayme Tiomno School in Theoretical Physics](#)
Minicourse on group theory in physics for first-year undergraduate students
- 2020  An Introduction to $\LaTeX$
[XV Oceanography Thematic Week](#)
Oceanographic Institute, University of São Paulo
- 2018  An Introduction to $\LaTeX$
[I Jayme Tiomno School in Theoretical Physics](#)

Teaching Assistant

- 2025  Topics in Celestial Holography
Institute of Physics, University of São Paulo
Graduate-level course
-  Advanced Topics in General Relativity
Institute of Physics, University of São Paulo
Graduate-level course
- 2024  Electromagnetic Phenomena
Federal University of ABC
Undergraduate-level course covering introductory electrodynamics
- 2021  Classical Mechanics II
Federal University of ABC
Undergraduate-level course covering analytical mechanics

Teaching Experience (continued)

- 2019  **Mathematical Physics I**
Institute of Physics, University of São Paulo
Undergraduate-level course covering introductory Fourier analysis

Participation in Panels

- 2026  **VI ICTP-SAIFR Summer School for Young Physicists**
Panel with graduate students in physics

Outreach and Miscellanea

-  **VI ICTP-SAIFR Summer School for Young Physicists**
Tutored a course on black hole physics for high school students
- 2025  **International Young Physicists Tournament (Brazil)**
Juror for the final stage of the 2025 edition of the IYPT Brazil
-  **V ICTP-SAIFR Summer School for Young Physicists**
Tutored a course on black hole physics for high school students
- 2018–  **Physics Stack Exchange**
Active contributor to the Physics Stack Exchange Q&A website
Most prolific contributor to the [qft-in-curved-spacetime](#) tag

Awards

- 2025  “Outstanding Recognition” award at the **UFABC Physics Week**
- awarded to 12 out of 45 participants based on talks about their research progress

Research Presentations

Invited Talks

- 2026  Sound as a gauge theory and its infrared triangle
- **Simon’s Collaboration on Celestial Holography Zoom Seminars** (10 Mar. 2026)
- 2025  The sky as a Killing horizon
- **Siembra-HoLAGrav Future of Physics** (08 May 2025)

Contributed Talks

- 2026  Sound as a gauge theory and its infrared triangle
- **Simons–Balseiro School and Workshop on Advanced Aspects of QFT** (Instituto Balseiro, 18–27 May 2026)
- 2025  Quantum Gravity and the Night Sky
- **UFABC Physics Week** (Federal University of ABC, 14–17 Oct. 2025)
- 2024  Positive mass in general relativity without energy conditions
- **XLVII Paulo Leal Ferreira Congress** (Institute of Theoretical Physics, São Paulo State University, 29 Oct.–01 Nov. 2024)
- 2023  Nonperturbative renormalization group flow for a particle detector
- **Golden Wedding of Black Holes and Thermodynamics** (online conference, 04–08 Dec. 2023)

Research Presentations (continued)

Poster Presentations

- 2026  You can teach gravitational wave physics using thermodynamics
- [Third generation gravitational wave detectors: The view from Latin America](#) (Principia Institute, 29 June–03 July 2026)
- 2026  We can use the renormalization group to study the nonperturbative behavior of particle detectors
- [Simons–Balseiro School and Workshop on Advanced Aspects of QFT](#) (Instituto Balseiro, 18–27 May 2026)
- 2025  Linear sound has asymptotic symmetries
- [XI QCD Meets Gravity Conference](#) (Principia Institute, 15–19 Dec. 2025)
 - [XI QCD Meets Gravity School](#) (Institute of Theoretical Physics, São Paulo State University, 08–12 Dec. 2025)
- 2025  Stars with extreme pressures can mimic black holes
- [XLVIII Paulo Leal Ferreira Congress](#) (Institute of Theoretical Physics, São Paulo State University, 14–17 Oct. 2025)
- 2025  Killing horizons naturally define asymptotic symmetries
- [First International Latin American Conference on Gravitational Waves](#) (National Institute for Space Research, 15–19 Sep. 2025)
 - [VI Siembra-HoLAGrav Young Frontiers Meeting at ICTP-SAIFR](#) (Institute of Theoretical Physics, São Paulo State University, 30 June–11 July 2025)
 - [Program on Anomalies, Topology and Quantum Information in Field Theory and Condensed Matter Physics](#) (Principia Institute, 16–27 June 2025)
- 2024  Negative masses are unstable and we don't need energy conditions to prove it
- [Witnessing Quantum Aspects of Gravity in a Lab](#) (Principia Institute, 23–27 Sep. 2024)
- 2023  Nonperturbative aspects of quantum field theory in curved spacetime
- [XLVI Paulo Leal Ferreira Congress](#) (Institute of Theoretical Physics, São Paulo State University, 24–27 Oct. 2023)
 - [V Jayme Tiomno Physics School](#) (Institute of Physics, University of São Paulo, 4–8 Sep. 2023)
 - [Interfaces Between Quantum and Classical Statistical Mechanics](#) (Institute of Mathematics and Statistics, University of São Paulo, 24–28 July 2023)

Flash Talks

- 2023–2024  Nonperturbative renormalization group flow for a particle detector
- [School on Phase Transitions and Gravitational Waves](#) (International Institute of Physics, Federal University of Rio Grande do Norte, 4–8 Mar. 2024)
 - [Quantum Spacetime and the Renormalization Group 2023](#) (Sant'Elmo Beach Hotel, 2–6 Oct. 2023)